

1 Solution — GIAM 1.1

Problem 7

1.1 Problem

Consider the process of long division. Does this algorithm give any insight as to why rational numbers have terminating or repeating decimal expansions? Explain.

1.2 The Answer — Yes

Long division *is* the proof. Watching the algorithm run on a rational a/b exposes the underlying mechanism: the remainder at each step lives in a *finite* set $\{0, 1, \dots, b-1\}$, so something must eventually happen — either the remainder hits 0 (and the expansion terminates) or it revisits a previous value (and the expansion cycles forever). There is no third option.

1.3 The Algorithm — A Quick Recap

To compute the decimal expansion of a/b (assume $0 < a < b$ for simplicity, so the integer part is 0):

1. Start with remainder $r_0 = a$.
2. Multiply by 10: $10 r_0$.
3. The next decimal digit is $d_1 = \lfloor 10 r_0 / b \rfloor$.
4. The next remainder is $r_1 = 10 r_0 \bmod b$.
5. Repeat with r_1 in place of r_0 : digit $d_2 = \lfloor 10 r_1 / b \rfloor$, new remainder $r_2 = 10 r_1 \bmod b$, and so on.

So the algorithm is governed by the **remainder iteration**

$$r_{k+1} = 10 r_k \bmod b,$$

with the emitted digit being $d_{k+1} = \lfloor 10 r_k / b \rfloor$.

1.4 The Key Observation — Finite State Space

At every step the remainder r_k is an integer with

$$0 \leq r_k < b.$$

So there are only b possible values r_k could take: $\{0, 1, 2, \dots, b-1\}$.

This is a *finite* set. As the algorithm produces an infinite sequence $r_0, r_1, r_2, \dots$ of remainders, **by the pigeonhole principle some value must repeat** — or the value 0 must eventually appear. There are only two ways for an infinite trajectory through a finite set to behave:

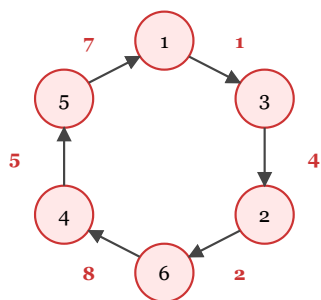
- **Hit 0**, in which case all subsequent remainders are 0 and all subsequent digits are 0: the expansion *terminates*.
- **Revisit a nonzero remainder**, in which case the trajectory has entered a cycle: from that point on the same sequence of remainders (and thus the same sequence of digits) repeats forever — the expansion is *eventually periodic*.

A *finite-state deterministic system* iterated forever can only terminate at an absorbing state (0) or enter a cycle. **There is no third behavior.**

That is the entire content of “every rational has a terminating or repeating decimal.” ■

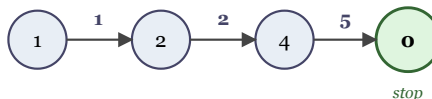
1.5 Visualisation — Cycle vs. Termination

$1/7$: remainders cycle through 6 states



Digits emitted around the cycle: 1, 4, 2, 8, 5, 7 $\rightarrow 1/7 = 0.142857$

$1/8$: remainder reaches 0, terminates



Digits emitted along the chain: 1, 2, 5 $\rightarrow 1/8 = 0.125$ (then all zero)

Figure 1.1: $1/7$'s remainders run through the cycle $1 \rightarrow 3 \rightarrow 2 \rightarrow 6 \rightarrow 4 \rightarrow 5 \rightarrow 1$, emitting the digits 1, 4, 2, 8, 5, 7 — the period of $1/7$. $1/8$'s remainders chain to 0 in three steps, emitting 1, 2, 5 — the terminating digits of $1/8 = 0.125$.

1.5.1 $1/7$ — The Cycle Case

Long division of $1 \div 7$:

step	remainder r_k	$10 r_k$	digit d_{k+1}	next r_{k+1}
0	1	10	1	3
1	3	30	4	2
2	2	20	2	6
3	6	60	8	4
4	4	40	5	5
5	5	50	7	$1 \leftarrow \text{back to start}$

Table 1.1

After 6 steps the remainder returns to 1. The cycle of remainders is 1, 3, 2, 6, 4, 5; the cycle of digits is 1, 4, 2, 8, 5, 7. From step 6 onward everything repeats:

$$1/7 = 0.\overline{142857}.$$

1.5.2 $1/8$ — The Terminating Case

Long division of $1 \div 8$:

step	remainder r_k	$10 r_k$	digit d_{k+1}	next r_{k+1}
0	1	10	1	2
1	2	20	2	4

2	4	40	5	0	$\leftarrow$ terminates
---	---	----	---	---	-------------------------

Table 1.2

Remainder 0 at step 3. After that every $r_k = 0$ and every $d_k = 0$. So

$$1/8 = 0.125 = 0.125000\dots$$

1.6 Why the Two Cases Match the Prime-Factor Rule

These two cases are *exactly* the dichotomy from Problem 5:

- The remainder hits 0 iff the long-division process exhausts the factors of 10 in b . This happens iff every prime factor of b (in lowest terms) is in $\{2, 5\}$.
- Otherwise, b has a prime factor coprime to 10, and multiplying by 10 never reduces that factor. The remainder iteration is a bijection on the subset of $\{1, \dots, b-1\}$ coprime to b , and never hits 0 — it must cycle.

So the prime-factor rule and the long-division rule are the same theorem, viewed from two sides:

Algebraic side	Algorithmic side
$b = 2^m 5^n \iff$ terminates	“remainder hits 0” $\iff$ terminates
b has a prime factor outside $\{2, 5\} \iff$ repeats	“remainders cycle without hitting 0” $\iff$ repeats

Table 1.3

Long division *exhibits* the algebraic condition by running it.

1.7 Remark — A Bound on the Period Length

The pigeonhole argument gives a *quantitative* bound: at most b steps before some remainder repeats. So the period of a/b is at most $b - 1$.

Sharper: when $\gcd(b, 10) = 1$, only nonzero remainders coprime to b can appear (and the iteration is a bijection on this set), so the period divides $\varphi(b)$ (Euler's totient). For prime b , this gives period dividing $b - 1$ — by Fermat's little theorem.

b	$\varphi(b)$	maximum period	actual period for $1/b$
7	6	6	$6 - 1/7 = 0.\overline{142857}$
11	10	10	$2 - 1/11 = 0.\overline{09}$
13	12	12	$6 - 1/13 = 0.\overline{076923}$
17	16	16	16 — long-period prime
19	18	18	18 — long-period prime

Table 1.4

When the period actually *equals* $\varphi(b)$ — as for $b = 7, 17, 19$ — we call b a **long-period prime** for base 10. (Equivalently: 10 is a primitive root modulo b .)

1.8 Remark — Pre-Period from 2s and 5s in the Denominator

If $b = 2^m 5^n \cdot b'$ with $\gcd(b', 10) = 1$ and $b' > 1$, the expansion has *both* a pre-period and a period:

- The first $\max(m, n)$ digits are the “pre-period” — they don't repeat. They are “consumed” by the 2s and 5s in the denominator.
- The remaining digits form the period, of length $\text{ord}_{b'}(10)$.

Example: $1/12 = 1/(2^2 \cdot 3)$. Here $\max(m, n) = 2$ and $b' = 3$, with $\text{ord}_3(10) = 1$. So $1/12$ has a 2-digit pre-period and a 1-digit period:

$$1/12 = 0.08\overline{3}.$$

Long division confirms it: remainders $1 \rightarrow 10 \rightarrow 4 \rightarrow 4 \rightarrow 4 \rightarrow \dots$ — the first two steps drain the 2s out of the denominator (giving the pre-period digits 0, 8), then the

remainder 4 sticks (a fixed point of the iteration $r \mapsto 10r \bmod 12$ for $r = 4$: $40 \bmod 12 = 4$).

1.9 Remark — The Converse: Periodic Decimal $\implies$ Rational

Long division proves *one direction* of “rational iff eventually periodic” — namely, that rationals have terminating-or-repeating expansions. The *converse* direction needs a separate argument:

If x has a decimal expansion with pre-period of length a and period of length p , then

$$10^{a+p}x - 10^a x$$

eliminates the repeating tail (the same infinite suffix appears on both sides). What remains is a *finite* difference of digit-segments — an integer. So

$$x(10^{a+p} - 10^a) \in \mathbb{Z},$$

making $x = N/(10^{a+p} - 10^a)$ for some integer N — a rational. ■

Combined with the long-division direction, this gives the full equivalence

$$x \in \mathbb{Q} \iff \text{decimal expansion of } x \text{ is eventually periodic.}$$

1.10 Remark — Generalisation to Other Bases

The long-division argument generalises immediately. In *any* base $B \geq 2$, the remainder at each step lies in $\{0, 1, \dots, b-1\}$, and the iteration is $r_{k+1} = B r_k \bmod b$. By pigeonhole the trajectory cycles or hits 0. So *every rational has a terminating-or-repeating expansion in every base*. The specific dichotomy “terminates iff $b = 2^m 5^n$ ” is base-10-specific; in base B , the condition is “ b ’s prime factors are all among those of B ”. (E.g., in base 6, terminates iff $b = 2^m 3^n$.)

The argument’s logical content is base-independent: bounded remainders + deterministic iteration = eventual cycle.

1.11 Remark — Long Division as a Finite-State Machine

A clean way to state the conclusion: long division of a/b is a **deterministic finite-state automaton** with state set $\{0, 1, \dots, b-1\}$ (the possible remainders), output alphabet $\{0, 1, \dots, 9\}$ (the digits), and transition rule $r \mapsto 10r \bmod b$ outputting $\lfloor 10r/b \rfloor$.

Every deterministic finite-state automaton, when run forever, either reaches a halting state (0 in our case) or enters a cycle. The output sequence is therefore either *finite* (the halting case $\rightarrow$ terminating decimal) or *eventually periodic* (the cycling case $\rightarrow$ repeating decimal).

This is exactly the dichotomy the problem asks us to explain — and long division is the algorithmic embodiment of the proof.